

Patent Vouchers as Innovation Incentives

Theory and Evidence from Europe

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This is a wonderful paper!

- Ambitious.
- Important.
- Timely.
- Combining theory and empirics in a thoughtful way.

What is it about?

Given a desired innovation reward, what is the cheapest way to pay it?

- Cash prize: reward financed through public funds.
- Transferable exclusivity voucher: rewarding an innovation by extending exclusivity on another product.
- Key insight: if post-patent competition is imperfect, the originator's willingness to pay for an extension can exceed the social gain from expiry due to business stealing effects. It is also easier to politically approve.

A transferable voucher can be better than a cash prize

- Cost-over-reward ratio (akin to MVPF):

$$\rho = \frac{\text{social surplus loss from extension}}{\text{reward delivered to innovator}} \equiv \frac{\text{social surplus loss from extension}}{\text{WTP by monopolist}}.$$

- Voucher better if loss in extension smaller than rents extracted from voucher, aka $\rho < 1$.
- This can happen if post-expiry rents go to generic firms rather than consumers/payers.
- The mechanism extracts WTP from existing patent holder, transfers it to innovator directly rather than generic companies.

A tension

Voucher works best when expiry destroys originator rents, but generic competition does not pass all those rents through to consumers/payers.

Is this a unicorn?



Empirically, they find it is not

Amazing prescription sales data for 15 countries from previous work, focus 2002-2012.

Steps:

- 1 Estimate voucher value from revenue preservation after loss of exclusivity.
- 2 Estimate demand by country and therapeutic class using price and quantity variation.
- 3 Compute cost-over-reward ratios for likely voucher users and countries.

Main empirical results

- Country level: $\rho_i < 1$ for all 15 countries.
- Union level: $\rho^U \simeq 0.76$, about 24% cheaper than cash.
- With GDP-based contributions: 8 of 15 countries favor vouchers, representing about 88% of population.

Point 1: Are the results conservative?

- The analysis (theory+empirics) is relatively conservative, although a bit on both ends.

Conservative

- all drug payments have a cost of public funds
- empirical estimates consider upper and lower bound of costs of drugs

Generous

- transferable voucher creates uncertainty (discussed with alternative interesting mechanism)
- assumption on perfect rent extraction at the auction stage ←

The paper is strongest as a comparison of financing instruments, conditional on society having chosen Γ .

Point 1: How Competitive Is The Voucher Market?

- The benchmark works best when the voucher auction is thick.

Many buyers

- bidders compete up the price
- innovator captures much of voucher value
- ρ close to benchmark.

Thin buyer market

- buyer captures rents
- social cost is still incurred
- reward transferred to innovator is lower.

Useful empirical object: you can exploit the gap between the first and second bidder.

Point 1: How Much Does The Second Bid Matter?

Year	Highest value	Second value	Second / first
2002	Plavix, 506	Lipitor, 503	0.99
2003	Plavix, 522	Lipitor, 518	0.99
2004	Plavix, 538	Lipitor, 534	0.99
2005	Plavix, 555	Lipitor, 551	0.99
2006	Plavix, 572	Lipitor, 568	0.99
2007	Plavix, 590	Lipitor, 586	0.99
2008	Plavix, 608	Lipitor, 604	0.99
2009	Lipitor, 622	Zyprexa, 363	0.58
2010	Lipitor, 642	Zyprexa, 374	0.58
2011	Lipitor, 661	Seroquel, 367	0.55

Values from Table 2, one-year voucher, million euros.

Critical for calculation

A factor of 0.55 could raise ρ above one, as it directly enters the denominator. Note: even if it can be reversed for some drugs, vouchers still have the political approval process in favor.

Point 2: Selection Into Voucher Use

- The voucher will be used where extension is most privately valuable.
- But those may be exactly the products where generic entry would have helped the most.

Empirical question

Do drugs that self-select into the mechanism also have higher ρ ?
Would this be expected from the theory based on fundamentals?

Point 3: Can We See The Access Margin?

- Most of the focus is on the competition effects/price bargaining (denominator).
- In the model, access depends on the price threshold (numerator):

$$\theta^* = \gamma p.$$

- A voucher keeps p high, so some patients remain untreated for longer than expected.
- It would be useful to show access effects to assess backlash potential.
 - E.g., how many patients are delayed, in which countries/classes, for what kinds of drugs?

Takeaways: a great paper!

- I learned a lot from the paper: vouchers deserve to be evaluated seriously, not dismissed mechanically.
- The framework gives us the right tools to discuss incidence and efficiency.

Three implementation questions

- 1 Can innovators reliably monetize the voucher?
- 2 Are selected extensions the ones where delayed competition is most costly?
- 3 What are the welfare losses in terms of health, which could raise fairness concerns?